

Communication Electronics Super Condensed Cheat Sheet

Waves & Antennas

- $\lambda = c/f$
(wavelength, used often in antenna problems)
- Quarter-wave antenna:
 $L = \lambda/4$
- Half-wave dipole:
 $L = \lambda/2$
- Propagation time:
 $t = d/v_p$
(exam problems: Earth-Mars delay, coax velocity factor)
 $v = c \cdot VF$

Decibels & Gains

- Voltage gain:
 $G_v(dB) = 20 \log_{10}(V_{out}/V_{in})$
- Current gain:
 $G_i(dB) = 20 \log_{10}(I_{out}/I_{in})$
- Power gain:
 $G_p(dB) = 10 \log_{10}(P_{out}/P_{in})$
- Cascaded:
 $G_{tot} = \prod A_n$
, in dB
 $G_{dB} = \sum G_{n,dB}$
- Power in dBm:
 $P_{dBm} = 10 \log_{10}(P/1mW)$

Exam tip: Gains in series → multiply (num), add (dB). Loss = negative dB.

RC Filters

- Corner frequency:
 $f_c = 1/(2\pi RC)$
- Transfer:
 $H(j\omega) = \frac{\omega_c}{j\omega + \omega_c}$
→ at
 f_c
: -3 dB point.
- Attenuation slope: -20 dB/decade per pole (exam Qs ask for slope extrapolation).

Resonance & RLC

- $X_L = 2\pi fL, X_C = 1/(2\pi fC)$
 - Resonance:
 $f_0 = 1/(2\pi\sqrt{LC})$
 - Bandwidth:
 $BW = f_0/Q, Q = f_0/BW$
- Exam tip:** Identify given L, C, R, then solve for f0, Q, BW quickly.

Filters (Butterworth / Chebyshev)

- $L_{ref} = R_0/(2\pi f_c), C_{ref} = 1/(2\pi R_0 f_c)$
- Butterworth: flat passband, smooth. Chebyshev: ripple but sharper cutoff.

Matching Networks

- $Q = X_L/R_s$
,
 $R_p = (Q^2 + 1)R_s$
- Series-to-parallel:
 $L_p = L_s(1 + 1/Q^2)$
- $C_p = 1/((2\pi f)^2 L_p)$

Exam tip: Shunt element always on higher impedance side. Match source→load for max power transfer.

Oscillators & AM

- Barkhausen:
 $|A\beta| \geq 1, \angle A\beta = 0^\circ$
(oscillation condition)
- AM:
 $s(t) = [1 + m \cos(\omega_m t)] \cos(\omega_c t)$
, modulation index
 $m = A_m/A_c$
Exam tip: If $m > 1$
→ overmodulation distortion. Oscillator generates carrier before modulation.

High-Value Exam Problem Reminders:

- Antenna length ↔ frequency:** Use $f = c/\lambda$
, then quarter or half λ .
- Cascaded stages:** If in dB → add; if numeric → multiply. Convert back/forth as needed.
- Filter cutoff:**
 $f_c = 1/(2\pi RC)$
and slope rules (−20 dB/decade per pole).
- Q-factor:**
 $Q = f_0/BW$
(fast way if f0 and BW given).
- Convert power levels:**
 P_{dBm}
↔ W using $10^{dBm/10}/1000$